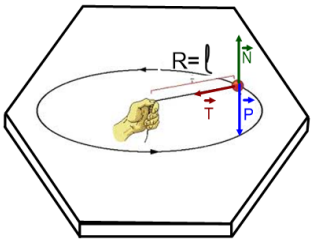
Componente Curricular: Física – Dinâmica
Professora: Rosana C. **Série:** 2ª **Data:**Agosto.
Conteúdo: Força Centrípeta - Teoria

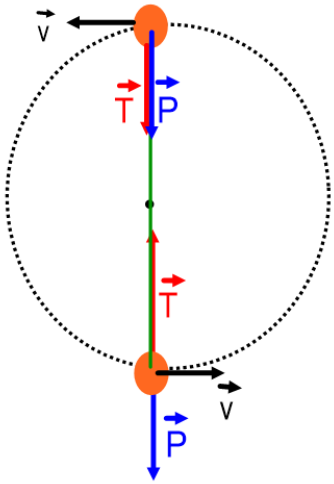
1- Corpo preso em um fio, movendo-se em um plano horizontal.



$$F_{cp} = T$$

$$F_{cp} = m.\frac{v^2}{R} \Rightarrow T = m.\frac{v^2}{R}$$

2- Corpo preso por um fio, movendo-se em um plano vertical.



Posição mais baixa

$$F_{cp} = T - P$$

$$T - P = m.\frac{v^2}{R}$$

Posição mais alta

$$F_{cp} = T + P$$

$$T + P = m.\frac{v^2}{R}$$

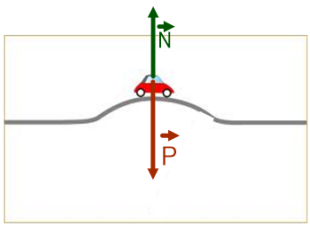
Cálculo da Velocidade Mínima

$v_{min} \Rightarrow T \to 0$ ♥ (posição mais alta)

♥ $P = m.\frac{v_{min}^2}{R} \Rightarrow m.g = m.\frac{v_{min}^2}{R}$

♥ $v_{min} = \sqrt{R.g}$

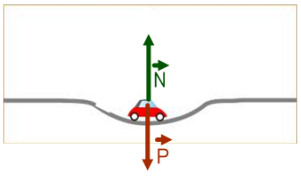
Lombada



P > N

$$P - N = m.\frac{v^2}{R}$$

Depressão

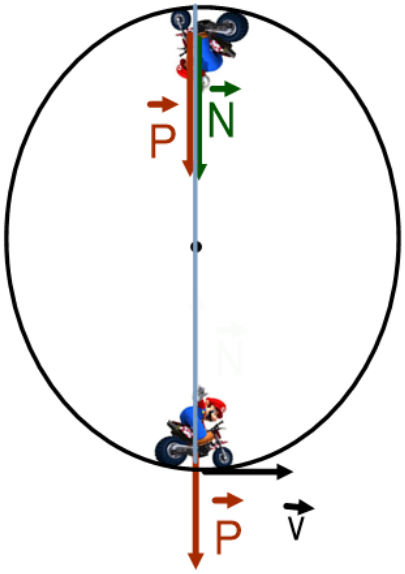


N > P

$$N - P = m.\frac{v^2}{R}$$

Globo da Morte

Posição mais baixa ♥ N > P



$$N - P = m.\frac{v^2}{R}$$

Posição mais alta

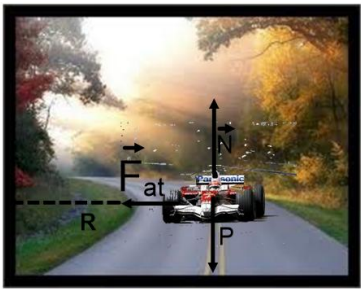
♥ $N + P = m.\frac{v^2}{R}$

♥ $v_{mín} \Rightarrow N \to 0$

♥ $P = m.\frac{v_{mín}^2}{R} \Rightarrow m.g = m.\frac{v_{mín}^2}{R}$

$$v_{mín} = \sqrt{R.g}$$

Automóvel - Curva

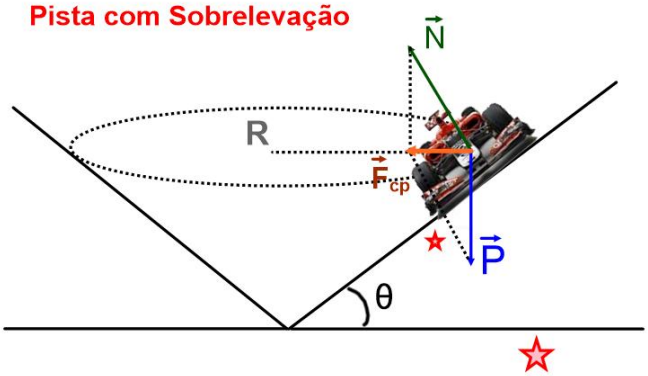


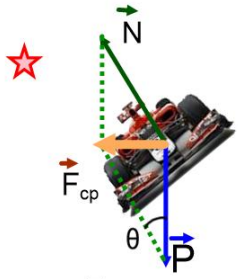
$$F_{cp} = F_{at} \Rightarrow F_{at} = m.\frac{v^2}{R}$$

$$F_{at} = m.\frac{v^2}{R} \Rightarrow \mu.m.g = m.\frac{v^2}{R}$$

☆ $\mu = \frac{v^2}{R.g}$

Pista com Sobrelevação



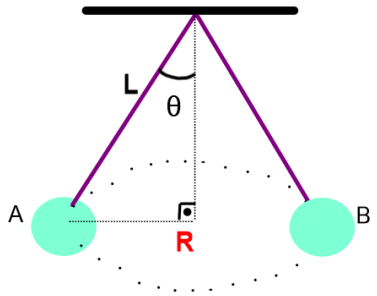




$$tg\theta = \frac{F_{cp}}{P}$$

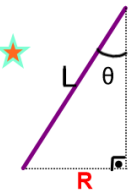
$$tg\theta = \frac{m.v^2}{R.m.g} \Rightarrow tg\theta = \frac{v^2}{R.g}$$

Pêndulo Cônico



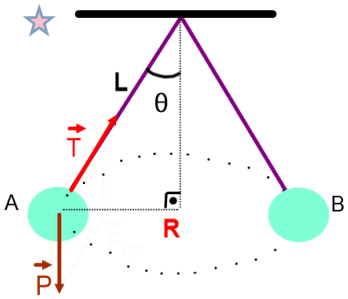
θ= ângulo de inclinação do fio com a vertical

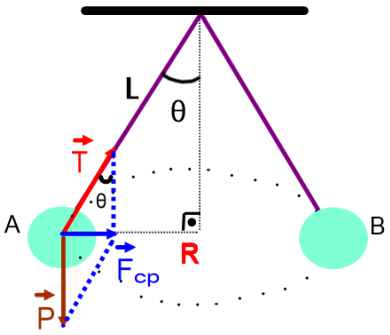
Δ de comprimentos



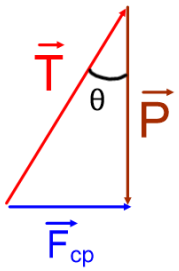
$$\text{sen}\theta = \frac{R}{L} \Rightarrow \boxed{R = L\text{sen}\theta}$$

Forças





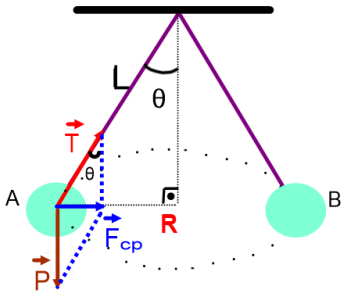
Δ de forças

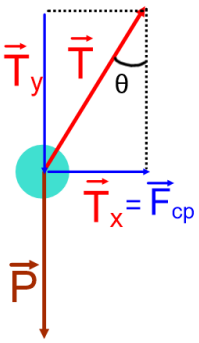


Determinação da forças

* Escrever senθ, cosθ, tgθ

Entendendo melhor





Vertical: repouso (F_R=0)

$T_y = T.\cos\theta$

$T_y = P$

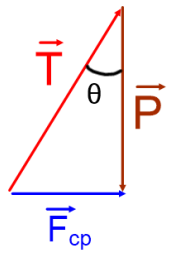
Horizontal (Radial)

$T_x = T.\text{sen}\theta$

$T_x = F_{cp}$

Período do Pêndulo

★ $\text{tg}\,\theta = \frac{F_{cp}}{P}$



★ $\frac{\text{sen}\,\theta}{\cos\theta} = \frac{m.\frac{v^2}{R}}{m.g} \Rightarrow \frac{\text{sen}\,\theta}{\cos\theta} = \frac{v^2}{R.g} \Rightarrow \frac{\text{sen}\,\theta}{\cos\theta} = \frac{\left(\frac{2\pi R}{T}\right)^2}{R.g}$

$\Rightarrow \frac{\text{sen}\,\theta}{\cos\theta} = \frac{\frac{4\pi^2 R^2}{T^2}}{R.g} \Rightarrow \frac{\text{sen}\,\theta}{\cos\theta} = \frac{4\pi^2 R^2}{T^2.R.g} \Rightarrow \frac{\text{sen}\,\theta}{\cos\theta} = \frac{4\pi^2 .l.sen\theta}{T^2.g}$

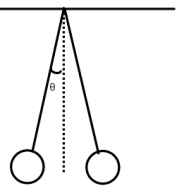
$\Rightarrow T^2 = \frac{4\pi^2 .l.\cos\theta}{g} \Rightarrow T = 2\pi\sqrt{\frac{l.\cos\theta}{g}}$

★ *Pequenas oscilações $\theta \to 0;\ \cos\theta \to 1$

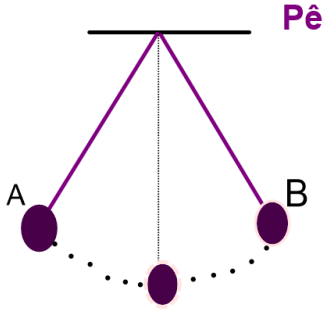
★ $T = 2\pi\sqrt{\frac{l}{g}}$

★ *O período do pêndulo não depende da massa da esfera

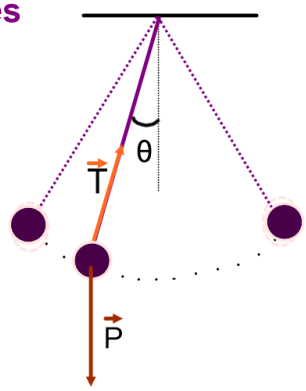
★

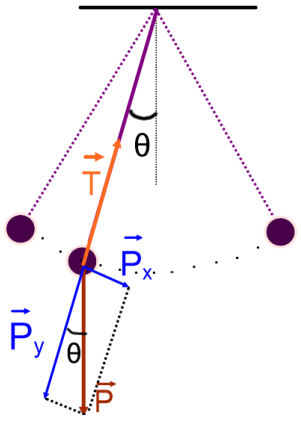


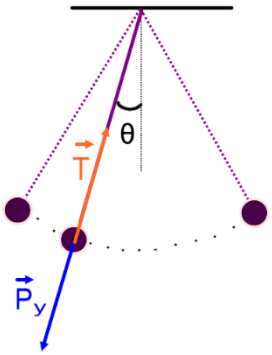
Pêndulo Simples

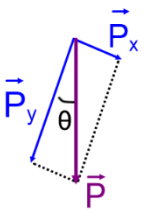


O pêndulo oscila no plano vertical, entre as posições A e B.







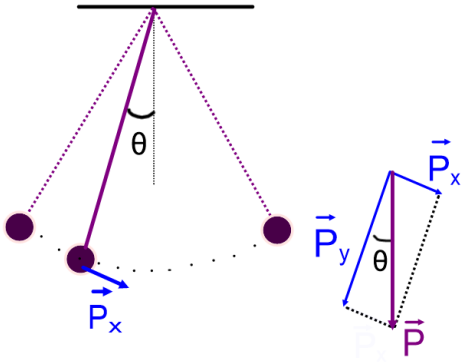


Força Centrípeta

* $F_{cp} = T - P_y$

* $T - P_y = m.\ a_{cp}$

* $P_y = P\ .\ \cos\theta$

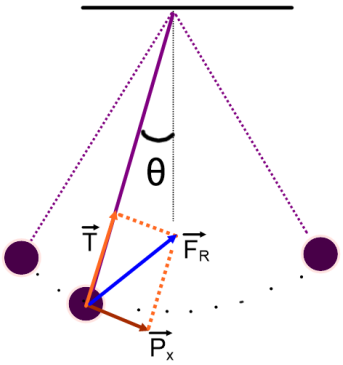


Força Tangencial

* $F_t = P_x$

* $P_x = m.\ a_t$

* $P_x = P\ .\ sen\theta$



Força Resultante

$$F_R = \sqrt{P_x^2 + F_{cp}^2}$$

O Pêndulo Simples descreve um Movimento Circular não Uniforme